

Algorithmic Pseudocode Sample 34

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1: procedure ADAPTIVEESTIMATION( $n, p_0, N, a_{LW}, t_{RS}$ )
2:   draw  $\{x_k\}$  from  $p_0(x)$ 
3:    $\{\omega_k\} \leftarrow \{1/K\}$ 
4:   for  $i \in 1..N$  do
5:      $\hat{T}_\chi \leftarrow \sum_j \omega_k \cdot x_k$ 
6:      $\tau \leftarrow \xi \cdot \hat{T}_\chi$ 
7:     for  $j \in 1..R$  do
8:        $m_j \leftarrow \text{EXPERIMENT}(\tau)$ 
9:     end for
10:     $r_i \leftarrow \sum_{j=1}^R m_j$ 
11:     $\{\omega_k\} \leftarrow \{\omega_k \cdot p(r_i | T_\chi, R)\}$ 
12:     $\{\omega_k\} \leftarrow \{\omega_k / (\sum_k \omega_k)\}$ 
13:    if  $1 / \sum \omega_k^2 < n \cdot t_{RS}$  then
14:       $\{x_k\} \leftarrow \text{RESAMPLE}(\{x_k\}, \{\omega_k\}, a_{LW})$ 
15:       $\{\omega_k\} \leftarrow \{1/K\}$ 
16:    end if
17:  end for
18: return  $\hat{T}_\chi$ 
19: end procedure
20:
21: procedure RESAMPLE( $\{x_k\}, \{\omega_k\}, a_{LW}$ )
22:    $\mu \leftarrow \sum_k x_k \cdot \omega_k$ 
23:    $\sigma^2 \leftarrow \sum_k x_k^2 \cdot \omega_k - \mu^2$ 
24:    $\mu' \leftarrow a_{LW} \cdot x_k + (1 - a_{LW}) \cdot \mu$ 
25:    $\{x_k\} \leftarrow \text{NORMAL}(\mu', \sigma^2)$ 
26: return  $\{x_k\}$ 
27: end procedure
```
